THOMAS ADEWUMI UNIVERSITY

Oko-Irese, Kwara State, Nigeria.

STUDENTS INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES) REPORT

A REPORT ON THE INDUSTRIAL TRAINING UNDERTAKEN AT SANFAANI SAAS

BY

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24/10MSS047

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A REPORT SUBMITTED TO THE DEPARTMENT OF MATHEMATICAL AND COMPUTING SCIENCES, THOMAS ADEWUMI UNIVERSITY, IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE COMPLETION OF THE STUDENTS INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES).

TRAINING PERIOD:

July 15 - October 15, 2026

DECLARATION

I, Adeoye Emmanuel Opeyemi, with matriculation number 24/10MSS047, hereby declare that this SIWES report is an account of the industrial training I undertook at Sanfaani Saas from July 15 to October 15.

I confirm that the report was prepared by me based on my practical experiences, activities performed, knowledge acquired, and observations made during the training period. All external sources consulted in preparing this report have been duly acknowledged.

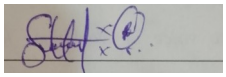
Student's Signature: _____

Date: _____

CERTIFICATION

This is to certify that this SIWES report was prepared by Adeoye Emmanuel Opeyemi, with matriculation number 24/10MSS047, of the Department of Mathematical and Computing Sciences, Thomas Adewumi University, following the completion of the approved industrial training programme.

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Signature:
Date: 15th October, 2026.



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Head of Department
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Date: _____

DEDICATION

This report is dedicated to God, whose grace, wisdom, and strength carried me through the entirety of this training period.

It is equally dedicated to my family, whose constant support provided a steady foundation during the demanding phases of this program. To my mentors, lecturers, and the colleagues who stood by me; thank you for your patience and for pushing me forward when the work got hard. Your guidance was invaluable in helping me translate classroom theory into real-world application.

My time during this attachment was more than just an opportunity to write code; it served as a strong reminder that technical skill means much more when it sits next to integrity, collaboration, and professional responsibility.

ACKNOWLEDGEMENTS

I thank God for the strength and clarity I needed throughout my SIWES program.

I am deeply grateful to the university's SIWES coordinators and my department for making this training period possible. This attachment gave me a chance to test classroom ideas against real software work. Navigating that gap between theory and practice taught me more than another semester of slides ever could.

I appreciate the management at Sanfaani Saas for bringing me into a working environment where I was allowed to ask questions, break things carefully, and figure out how to fix them. A special thank you goes to my industry supervisor, Mr. Taofeek, and the rest of the development team. Your patient corrections, thorough code reviews, and honest feedback fundamentally changed how I approach problem-solving.

Thank you to my family and friends for your constant encouragement, especially during the long hours spent debugging and learning new tools.

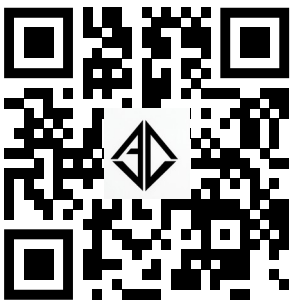
Finally, I want to thank the wider developer community and the maintainers of the open documentation I leaned on every day, including MDN, React, Node.js, and MongoDB. Good docs save hours. This experience showed me that strong engineers keep learning, and the best ones leave clearer notes for the next person.

ABSTRACT

This report covers my SIWES attachment at Sanfaani Saas, where I worked as a Junior Software Engineer Intern from July 15 to October 15. The work was mostly full-stack: reading existing code, connecting frontend and backend pieces, fixing configuration issues, testing features, and getting applications ready to deploy. The main project was SanfaaniService (SANFAANI), an internal operations console for charging sessions, workspace visits, customer records, inventory, sales, and receipts. Before the system, those records lived in separate manual processes, so it was hard to see what was happening across the business. We built one place where each action can leave a service record, transaction, receipt, and history trail. The live documentation for that work is at https://sanfaaniservicesdoc.vercel.app/. I also worked on Nexos, a productivity and project-management application, and on the SANFAANI documentation site itself. Across these tasks I used React, Vite, TypeScript, Tailwind, Node.js, Express, MongoDB, Supabase Auth, Git, Linux, and tools like WebStorm. The training improved my debugging habits, documentation discipline, and how I think about frontend behaviour, backend logic, data, and deployment as one system rather than separate subjects.



https://sanfaaniservicesdoc.vercel.app/.



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LIST OF ABBREVIATIONS

SIWES - Students Industrial Work Experience Scheme

ITF - Industrial Training Fund

HTML - HyperText Markup Language

CSS - Cascading Style Sheets

JS - JavaScript

API - Application Programming Interface

OS - Operating System

CHAPTER ONE
INTRODUCTION

1.1 Background of SIWES

SIWES is the practical training scheme that puts tertiary students into workplaces related to their courses. For a software engineering student, it is the bridge between lecture-hall concepts and how software is actually planned, built, tested, documented, and shipped. In class you learn programming, databases, networking, and software principles as separate topics. At work, a feature that looks simple to a user can touch interface design, frontend state, backend logic, database writes, authentication, validation, Git history, and deployment all at once. Seeing that chain in person changed how I read code.

1.2 Objectives of SIWES

For my own training, the objectives are:
- Apply classroom knowledge to real product requirements and codebase work.
- Gain practice with current tools, frameworks, databases, and collaborative workflows.
- Improve communication, time management, and professional discipline.
- Build confidence working inside existing repositories instead of only isolated exercises.
- Spot areas for continued growth in backend work, UX, and secure system design.

1.3 Scope of the Training

The training covered full-stack web development, API integration, and technical documentation. It involved building React interfaces, writing Express backend routes, managing MongoDB databases, and using Git for version control. The scope also included deploying applications to production environments and presenting technical concepts to junior team members.

1.4 Duration of the Training

The industrial training commenced on July 15 and concluded on October 15, lasting three months.

1.5 Significance of the Training

The program showed me how software engineering works outside a classroom. It reinforced the importance of maintainable code, clear documentation, and working within a team. The hands-on experience debugging live applications and managing deployments gave me confidence that academic theory alone cannot provide.

CHAPTER TWO
PROFILE OF THE ORGANISATION

2.1 Introduction

This chapter outlines the structure and operations of Sanfaani Saas, the host organization for my SIWES attachment, focusing on the software development unit where I was assigned.

2.2 History and Background of the Organisation

Sanfaani Saas is a technology company focused on software products and service-management tools. My attachment sat inside software development work: building and improving digital systems that support day-to-day business operations. The environment pushed a product way of thinking. Code was never only code. I had to ask who uses the screen, what data moves through it, whether the workflow holds up under real use, and whether another developer can read the repository without guessing.

2.3 Location and Address

Sanfaani Saas operates as a modern technology firm with distributed teams in Cameroon and office space in Ilorin, Kwara, Nigeria.

2.4 Vision and Mission

The company aims to simplify business operations through reliable software. Their mission is to build digital tools that replace fragmented manual processes with unified, traceable systems.

2.5 Organisational Structure

The working structure most relevant to my attachment looked like this:

Management/Operations -> Technical / Product Lead -> Software Development -> Product and Quality Support.

As an intern I sat close to the development workflow. I received tasks, studied existing patterns, made changes under guidance, tested the result, and reported blockers.

2.6 Departments or Units

The organization is divided into several units including Software Development, Information Technology, Product Support, and Operations. These units collaborate to ensure that software tools meet the actual needs of the business.

2.7 Functions and Services

Sanfaani Saas builds internal and customer-facing software. This includes operations consoles for managing workspace visits, charging sessions, inventory counts, and digital receipts.

2.8 The Unit of Attachment

Most of my time sat in the software development and product engineering workflow. The unit mixed frontend work, backend integration, database interaction, debugging, testing, version control, and attention to user experience. On the frontend I worked with components, forms, application state, routing, and responsive behaviour. On the backend I met API routes, request handling, validation, and authentication concepts. Overall, the unit matched what I want long term: to become a software engineer who treats interfaces, backend logic, systems, security, and product quality as connected responsibilities.

CHAPTER THREE
ACTIVITIES AND PRACTICAL EXPERIENCE

3.1 Introduction

This chapter details the technical tasks I performed, the tools I used, and the methodology I applied during my attachment.

3.2 Orientation and Familiarisation

During my first week, I was introduced to the company's repository structure, deployment environments, and communication channels. I received access to the codebase and learned the team's rules for creating branches, submitting pull requests, and handling environment variables safely.

3.3 Weekly Summary of Activities

Week	Activities Undertaken	Skills Acquired
Week 1-2	Orientation, environment setup, and codebase familiarisation.	Environment configuration, Git branching.
Week 3-4	Frontend UI work on SanfaaniService using React and Tailwind.	Component structuring, responsive design.
Week 5-6	Backend API integration with Express and MongoDB.	API consumption, database querying.
Week 7-8	Writing technical documentation for SanfaaniService.	Technical writing, diagramming flows.
Week 9-10	Work on Nexos project UI and application state management.	State management, user experience design.
Week 11	Lecturing newbies on Web Basics, HTML/JS, and Git.	Technical communication, mentoring.

Week 12	Final deployment checks, debugging, and report writing.	Production checks, system troubleshooting.

3.4 Detailed Description of Major Tasks

Project: SanfaaniService

SanfaaniService is an internal console for charging, workspace visits, customer records, inventory, sales, and receipts. The business used separate manual records for these operations. We built one system where each action records the service, transaction, receipt, and history in the same place. I read the frontend and backend codebase, fixed configuration problems, checked API connections, worked through database connectivity, and verified behaviour locally and in deployment.

Project: Sanfaani Documentation Site

I built and maintained the documentation site https://sanfaaniservicesdoc.vercel.app

The site is organised around why the platform was built, what it does, how the architecture hangs together, and what I learned while building it. Writing the docs forced me to mark what was implemented versus what was only designed.

Project: Nexos

Nexos is a productivity and project management application. I paid attention to information structure, how users move through screens, and whether status and priority were obvious without a lecture. The work improved my sense of reusable components, application state, and route organisation.

Task: Lecturing and Onboarding Newbies

As part of my role, I ran presentation sessions for new interns and beginners joining the tech team. I walked them through web fundamentals, explaining how HTML structures a page, how CSS styles it, and how JavaScript adds behavior. I also taught them the basics of version control with Git and GitHub.

We covered how to clone repositories, create branches, stage changes, and push their first commits without breaking the main branch. Teaching these concepts forced me to organize my own knowledge and communicate technical ideas clearly to a non-technical audience.

3.5 Tools, Equipment, and Software Used

- Frontend: HTML, CSS, JavaScript, React, TypeScript, Vite, Tailwind.
- Backend: Node.js, Express.
- Database / Auth: MongoDB, Mongoose, Supabase Auth.
- Dev environment: WebStorm, terminal, pnpm.
- Version control: Git and Github.
- OS: Linux (Kubuntu, KDE).
- Deployment: Vercel (frontend), Render (backend).

3.6 Application of Classroom Knowledge

Concepts from my university programming courses helped me understand the logic behind the application loops. My database management classes made it easier to grasp MongoDB schemas and data persistence. Furthermore, software engineering principles regarding modularity and clean code directly translated to how we structured React components.

3.7 Professional and Workplace Skills Acquired

I developed clearer communication habits, especially when reporting blockers or explaining code during reviews. I learned time management by breaking large features into smaller, testable commits. The environment also reinforced the importance of workplace discipline, showing up consistently, and taking ownership of assigned tasks.

CHAPTER FOUR
SKILLS ACQUIRED, CHALLENGES, AND SOLUTIONS

4.1 Technical Skills Acquired

Before the training my strongest base was HTML, CSS, JavaScript, React, and basic Node.js. By the end I was comfortable with TypeScript, Vite application structure, Express APIs, MongoDB connectivity, Git workflows, and Linux terminal troubleshooting. I also developed strong documentation habits.

4.2 Professional Skills Acquired

I improved my patience, ownership, and willingness to check assumptions before writing code. I learned how to read systems I did not start from scratch and how to improve software in small, reliable steps.

4.3 Challenges Encountered

Most of the useful lessons came from problems that were not obvious at first glance. Missing or malformed environment variables stopped services from starting. Development servers sometimes tried to bind a port already in use. MongoDB connections failed because of network rules or access configuration. Furthermore, working across repositories and remotes required careful version-control discipline.

4.4 Solutions to the Challenges

To address environment issues, I learned to verify required variables and compare local configuration with deployment requirements. When ports conflicted, I checked running processes and restarted on the correct port. For database failures, I traced the connection string and runtime logs instead of changing unrelated code. I became more deliberate about checking branch status before pushing to Git.

4.5 Relationship Between Theory and Practice

The attachment bridged the gap between theory and practice. In class, topics like authentication, API routing, and state management are taught in isolation. In the workplace, I saw how a single user login action touches the

interface, the API, the database, and the security configuration simultaneously.

4.6 Overall Assessment of the Training

The training was highly effective. It exposed me to professional standards, modern development workflows, and the practical realities of deploying software to actual users.

CHAPTER FIVE
SUMMARY, CONCLUSION, AND RECOMMENDATIONS

5.1 Summary

During my three months at Sanfaani Saas, I worked as a Junior Software Engineer Intern focusing on full-stack web development. I contributed to the SanfaaniService console, built its documentation site, and worked on the Nexos project. I also trained new team members on web basics and version control.

5.2 Conclusion

SIWES gave practical meaning to ideas that can feel separate in class. Interfaces, backend logic, databases, configuration, testing, Git, deployment, and user experience showed up as one job. I leave more interested in full-stack and backend engineering, with a clearer respect for Linux, secure habits, and writing that other people can follow. The engineer I want to become is dependable, curious, careful with security, and able to ship products that stay useful after the demo ends.

5.3 Recommendations

A. Recommendations to the Organisation
Host organisations should continue giving interns real but scoped tasks, with clear expectations and regular technical feedback.

B. Recommendations to Thomas Adewumi University
Institutions should prepare students with practical Git, API, database, Linux, testing, and deployment exercises before attachment. Documentation and communication should count as engineering work in academic grading.

C. Recommendations to Students
Students should keep a weekly log of problems, fixes, commands, and lessons; those notes make the final report honest. Interns should test complete user flows, not only single file changes, so integration bugs show up early.

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